

Visualizing Housing Market Trends

An Analysis of Sale Prices and Features using Tableau

Final Project Report

Submitted as part of the SkillWallet Internship Program

Team Members:

K. Naga Sai Parnika

A. Nityasri Sai Kalyani

K. Srujana

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1. Introduction

1.1 Project Overview

Prospective home buyers, home sellers, and real estate analysts frequently struggle to understand what actually drives house prices in a given market. Raw housing data — sale prices, square footage, bedrooms, bathrooms, condition, renovation history, and location — is often available, but it is rarely presented in a way that non-technical stakeholders can interpret. This project addresses that gap by building an interactive Tableau dashboard that visualizes approximately 21,609 records of King County-style residential housing data.

The workbook takes the raw dataset (Transformed_Housing_Data2.csv) through cleaning and preparation, then presents it through four coordinated visualizations: a KPI summary table, a renovation-based sales histogram, a house-age-versus-renovation-status pie chart, and a bedroom/bathroom/floor-based house age distribution chart. These are combined into a single 2x2 floating dashboard titled "Comprehensive House Data Analysis" and supported by a three-point guided storyboard that walks viewers through the key insights.

1.2 Purpose

The purpose of this project is to help two primary user groups make more confident, data-backed decisions:

- Home buyers, who want to evaluate whether a listed sale price is fair given a home's bedrooms, bathrooms, square footage, condition, and age, without manually comparing scattered listings.
- Real estate analysts, who want a clear, visual breakdown of how renovation history and house age influence final sale prices, without manually filtering and cross-referencing a dataset of over 21,000 records.

By combining data cleaning, visualization, dashboarding, and a guided narrative into one Tableau Public workbook, the project turns a large, unstructured dataset into an accessible decision-support tool.

2. Ideation Phase

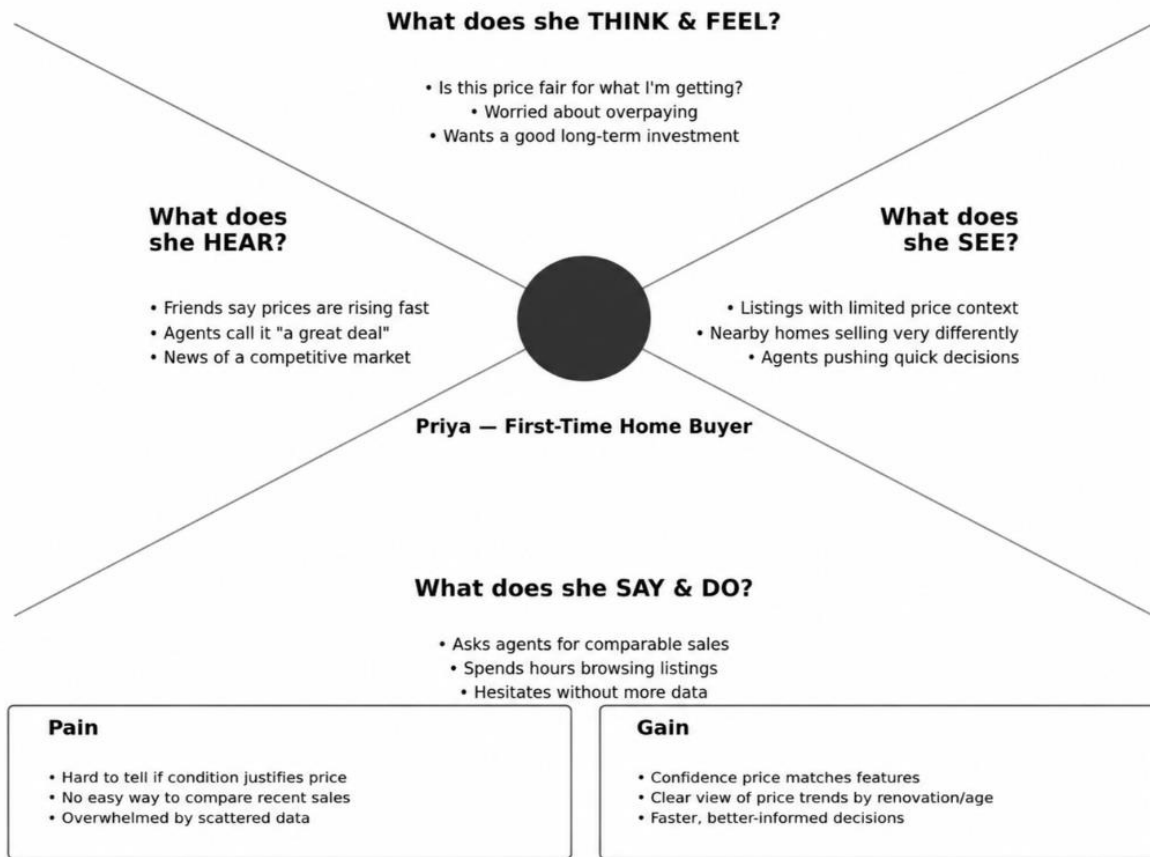
2.1 Problem Statement

Two customer problem statements were defined to frame the project from the perspective of its two primary user groups.

	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A prospective home buyer researching properties in the King County area	Evaluate whether a listed sale price is fair given the home's features (bedrooms, bathrooms, square footage, condition, and age)	I cannot easily compare a property against similar homes that have recently sold	Housing sale data is scattered across multiple sources and is not organized by comparable features	Uncertain and anxious about overpaying
PS-2	A real estate analyst advising clients on pricing strategy	Understand how renovation history and house age influence final sale prices	I don't have a clear, visual breakdown of sales trends by renovation status and age group	The raw housing dataset (21,609+ records) is too large to interpret manually and isn't summarized into digestible insights	Frustrated by the time spent manually filtering and cross-referencing spreadsheets

2.2 Empathy Map Canvas

An empathy map was built for "Priya," a persona representing a first-time home buyer using the dashboard, to capture what the target user thinks, feels, sees, hears, says, and does.

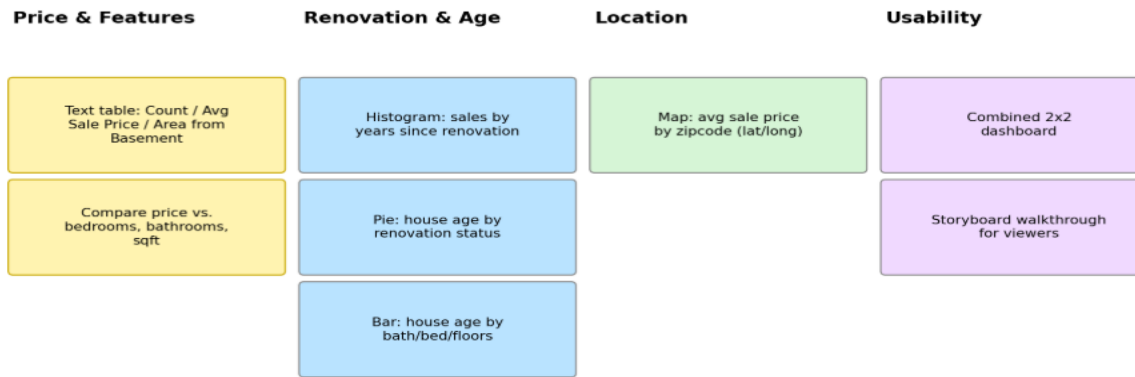


Key pain points identified were the difficulty of telling whether a home's condition justifies its price, the lack of an easy way to compare recent sales, and the feeling of being overwhelmed by scattered data. The corresponding gains the solution targets are confidence that price matches features, a clear view of price trends by renovation and age, and faster, better-informed decisions.

2.3 Brainstorming

The team began brainstorming from the selected problem statement: "How might we help home buyers and real estate analysts understand what drives housing sale prices, so they can make confident, well-informed decisions?"

Ideas were grouped into four themes — Price & Features, Renovation & Age, Location, and Usability — and then placed on an importance-versus-feasibility grid.



Idea Prioritization Grid — Importance vs. Feasibility



Idea	Importance	Feasibility
Text table: Count / Avg Sale Price / Area from Basement	High	High — built (Sheet 1)
Histogram: Total Sales by Years Since Renovation	High	High — built (Sheet 2)
Pie chart: House Age by Renovation Status	High	High — built (Sheet 3)
Bar chart: House Age by Bathrooms/Bedrooms/Floors	High	High — built (Sheet 4)
Map view: Avg Sale Price by Zipcode	Medium	Low — deferred, time constraints

3. Requirement Analysis

3.1 Customer Journey Map

The journey map below traces a real estate analyst or stakeholder exploring the dashboard, from first access through to a final pricing or marketing decision.

Stage	Access Dashboard	Review KPIs	Analyze Visualizations	Review Story	Make Decisions
Experience	Opens the Tableau workbook / published dashboard link	Scans KPI cards for record count, average sale price, and basement area	Filters and drills into charts on renovation, age, and room counts	Walks through the guided Tableau story	Applies insights to pricing, marketing, or investment decisions
Positive moments	Loads quickly with a clean, uncluttered layout	KPI cards make overall performance easy to grasp	Filters reveal patterns not obvious in raw data	Story ties numbers into a clear narrative	Feels confident the decision is backed by real data
Negative moments	Unsure where to start among several visualizations	Some KPIs lack context on their own	Too many filter combinations can feel overwhelming	Story can feel static without the live dashboard	Hesitates to act without a second opinion
Opportunities	Add a short "how to read this dashboard" note	Add benchmark context next to each KPI card	Suggest default filter presets	Refresh the story periodically	Add an exportable summary of key findings

3.2 Solution Requirement

Functional requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement
FR-1	Data Collection & Extraction	Connect housing dataset (CSV) to Tableau; verify column data types and row count

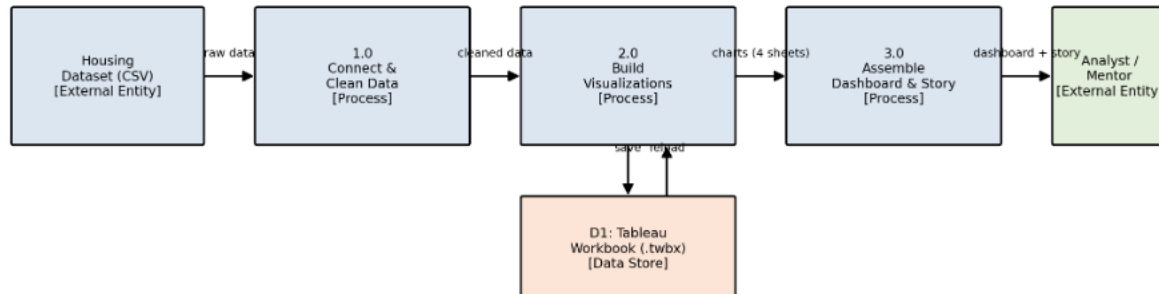
FR No.	Functional Requirement (Epic)	Sub Requirement
FR-2	Data Preparation	Check dataset for null values; check dataset for duplicate records
FR-3	Data Visualization	Build KPI summary sheet; sales-by-renovation chart; house age pie chart; bed/bath/floor age chart
FR-4	Dashboard Creation	Combine all visualizations into one dashboard; add interactive filters; format layout for readability
FR-5	Storyboard Creation	Build guided story with multiple points; add captions explaining each insight
FR-6	Performance Testing	Document amount of data loaded, filters used, and number of visualizations
FR-7	Documentation & Demonstration	Record end-to-end explanation video; write step-by-step project documentation

Non-functional requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Easy to navigate with clear titles and labels so non-technical viewers can interpret insights without training
NFR-2	Security	Dataset and workbook shared only through controlled links (GitHub, Tableau Public); no sensitive personal data exposed
NFR-3	Reliability	Visualizations consistently display values matching the underlying dataset every time the workbook is opened
NFR-4	Performance	Dashboard loads and responds to filter changes within a few seconds, given the ~21,600-row dataset
NFR-5	Availability	Published Tableau Public dashboard and GitHub repository remain accessible at all times for review
NFR-6	Scalability	Dashboard structure can accommodate additional visualizations or a larger dataset without a full redesign

3.3 Data Flow Diagram

The diagram below shows how data moves through the project: from the raw housing dataset, through cleaning and visualization in Tableau, into the final dashboard and story, and out to the analyst/mentor as the end user.

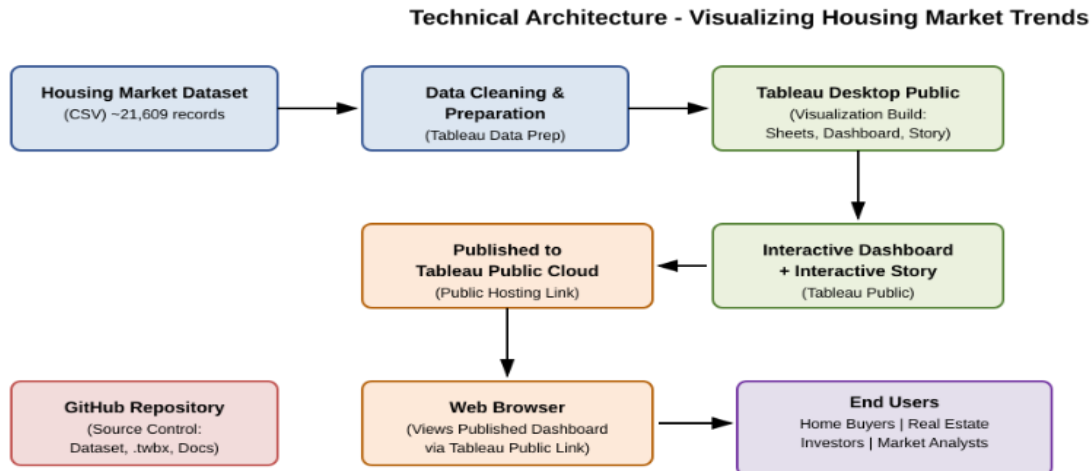


Selected user stories driving this flow are summarized below.

User Story No.	Epic	User Story	Priority
USN-1	Data Collection & Extraction	Connect the housing dataset (CSV) to Tableau so visualizations can begin	High
USN-2	Data Preparation	Verify the dataset for nulls and duplicates for accurate, clean data	High
USN-3	Data Visualization	Create charts showing sale price against house features to identify pricing patterns	High
USN-4	Dashboard	Combine visualizations into one dashboard for interactive stakeholder exploration	High
USN-5	Storyboard	Create a guided story so non-technical viewers can follow key insights	Medium
USN-6	Performance Testing	Document data volume, filters, and visualization count for transparency	Medium
USN-7	Documentation & Demo	Record a demo video and write documentation for end-to-end review	High
USN-8	Project Review	Mentor views published dashboard, story, and demo video to evaluate completeness	High

3.4 Technology Stack

The architecture flows from the raw CSV dataset, through data cleaning and preparation, into Tableau Desktop Public for visualization, dashboard, and story construction, and finally out to Tableau Public Cloud and a GitHub repository for sharing with end users.



#	Component	Description	Technology
1	User Interface	How stakeholders view the finished dashboard and story	Tableau Public (Web View), Web Browser
2	Application Logic-1	Cleans and prepares the raw housing CSV prior to visualization	Tableau Data Prep / Excel
3	Application Logic-2	Builds the individual visualizations, dashboard, and storyboard	Tableau Desktop Public
4	Application Logic-3	Publishes the finished dashboard and story for shareable viewing	Tableau Public (Publish/Share)
5	Database	Not applicable — data is sourced directly from a CSV file	Not Applicable
6	File Storage	Storage of the raw/cleaned dataset and Tableau workbook	Local File System (CSV, .twbx), Google Drive
7	Infrastructure	Local development; dashboard hosted on Tableau Public; version control via Git	Local Machine, Tableau Public, GitHub

4. Project Design

4.1 Problem Solution Fit

Customer segments: home buyers, home sellers, and real estate market analysts/investors evaluating King County-style residential properties.

Jobs-to-be-done: understand what actually drives a house's sale price, judge whether renovation history affects value, and compare price against age, size, condition, and room count without manually processing raw data.

Available alternatives — manually reviewing spreadsheets, browsing individual listing sites, relying on an agent's opinion, or building one-off Excel charts — all lack an aggregate, visual view of price-driving factors across the market. This is compounded by the underlying root cause: the raw dataset of roughly 21,600 records is large and unstructured for non-technical users, and no easy, centralized visual tool combines price, renovation, age, and feature data in one place.

The proposed solution is an interactive Tableau dashboard and storyboard that combines a KPI summary, a renovation-based sales histogram, a house-age-by-renovation pie chart, and a bedroom/bathroom/floor age bar chart into one easy-to-navigate view, published free via Tableau Public — moving users from a state of confusion and being overwhelmed by scattered raw data to one of informed confidence in pricing or negotiation decisions.

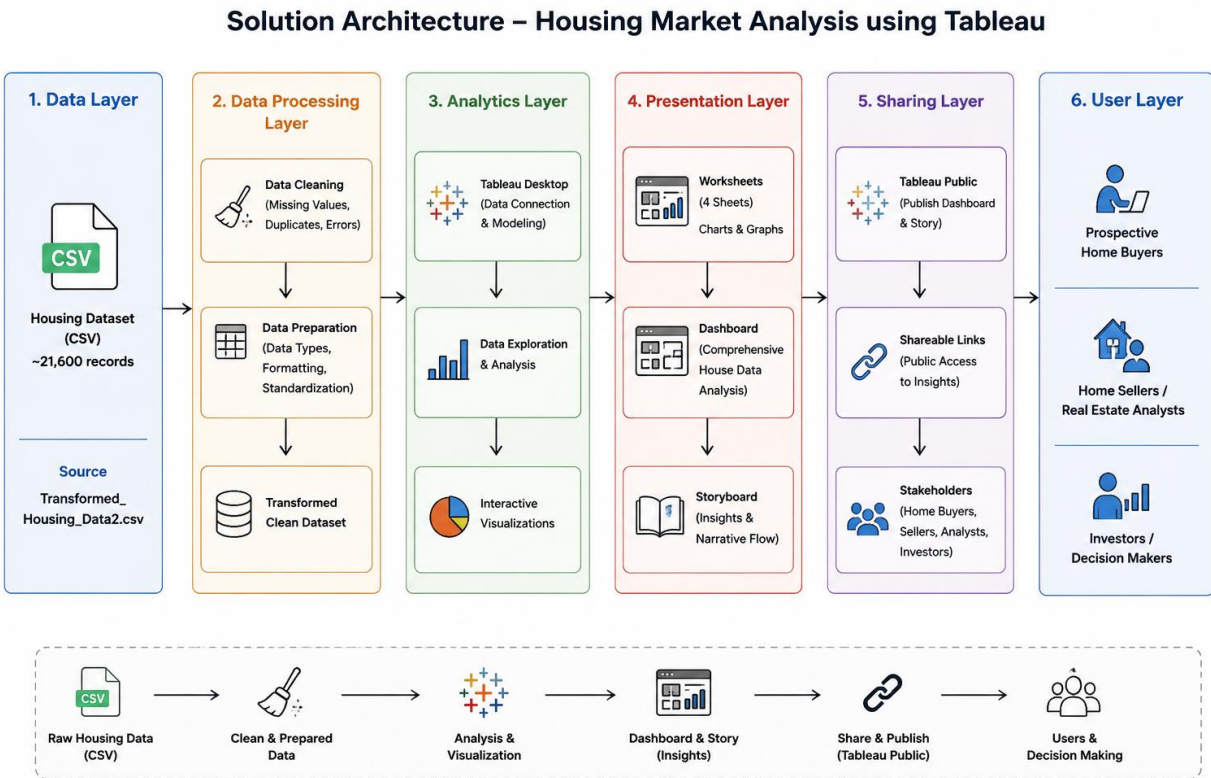
4.2 Proposed Solution

Parameter	Description
Problem Statement	Non-technical stakeholders struggle to interpret raw housing data (price, size, rooms, condition, renovation, location) to identify pricing patterns and the impact of renovation or age on value.
Idea / Solution	An interactive Tableau dashboard visualizing ~21,600 records via a KPI summary table, a renovation-based sales histogram, a house-age-vs-renovation pie chart, and a bed/bath/floor age chart, combined into a 2x2 dashboard with a guided storyboard.
Novelty / Uniqueness	Combines multiple related dimensions (renovation history, house age, room configuration, basement area) into one cohesive dashboard rather than isolated reports, with a guided narrative layer for non-technical viewers.
Social Impact	Helps buyers and sellers make more informed pricing decisions and gives analysts a quick, shareable reference tool instead of rebuilding analysis from raw data each time.
Business Model	Published free via Tableau Public for this internship project; a real-world extension could offer a freemium model with zipcode-level drill-downs or predictive pricing for agencies.

Parameter	Description
Scalability	The modular Tableau workbook can incorporate additional sheets, filters, or a larger/updated dataset without redesign, and could extend to other regions or a live data feed.

4.3 Solution Architecture

The solution architecture is organized into six layers: a Data Layer (the raw CSV, ~21,600 records), a Data Processing Layer (cleaning, preparation, and a transformed clean dataset), an Analytics Layer (Tableau Desktop, exploration, and interactive visualizations), a Presentation Layer (worksheets, dashboard, storyboard), a Sharing Layer (Tableau Public, shareable links, stakeholders), and a User Layer (home buyers, sellers/analysts, investors/decision makers).



5. Project Planning & Scheduling

5.1 Project Planning

Work was organized into two sprints covering data collection/preparation and visualization/dashboard/story delivery.

Sprint	Epic	User Story	Points	Priority	Owner
1	Data Collection & Extraction	Connect the housing dataset (CSV) to Tableau	2	High	K. Parnika
1	Data Collection & Extraction	Verify column data types and row count	1	High	A. Nityasri
1	Data Preparation	Check the dataset for null values	3	High	A. Nityasri
1	Data Preparation	Check the dataset for duplicate records	3	Medium	K. Srujana
1	Data Preparation	Format/create necessary fields for analysis	3	Medium	K. Srujana
2	Data Visualization	View a KPI summary (count, avg price, basement area)	2	High	K. Parnika
2	Data Visualization	View a histogram of sales by years since renovation	2	High	K. Parnika
2	Data Visualization	View a pie chart of house age by renovation status	2	High	K. Parnika
2	Data Visualization	View a bar chart of house age by bed/bath/floors	4	Medium	A. Nityasri
2	Dashboard	View all visualizations combined in one dashboard	5	High	A. Nityasri
2	Story	Follow a guided storyboard explaining key insights	5	High	K. Srujana

Sprint tracker:

Sprint	Total Points	Duration	Start Date	Planned End	Completed	Actual Release
Sprint-1	12	6 Days	23 June 2026	28 June 2026	12	28 June 2026
Sprint-2	20	6 Days	01 July 2026	06 July 2026	20	06 July 2026

With a 10-day sprint duration and team velocity of 20 points per sprint, the average velocity (AV) per iteration unit is $AV = \text{sprint duration} \div \text{velocity} = 20 \div 10 = 2$ story points per day. Both sprints were completed on schedule, with all planned story points delivered by their planned end dates.

6. Functional and Performance Testing

6.1 Performance Testing

Parameter	Result
Data Rendered	~21,609 housing records rendered successfully, covering Sale Price, Bedrooms, Bathrooms, Sqft Living, Sqft Basement, Condition, Year Built, Year Renovated, Zipcode, and Latitude/Longitude
Data Preprocessing	Dataset checked for null values and duplicate records; data types validated for all fields; inconsistent/invalid values standardized; calculated fields created for analysis
Utilization of Filters	Interactive filters applied to allow exploration by renovation status, house age range, and bedroom/bathroom/floor count
Calculated Fields Used	Years Since Renovation, House Age, Average Sale Price, Basement Area, House Age Grouping
Dashboard Design	4 visualizations (KPI summary, histogram, pie chart, bar chart) combined in a 2x2 floating layout titled "Comprehensive House Data Analysis"
Story Design	3 story points: House Age Distribution by Bathrooms/Bedrooms/Floors, Distribution of House Age by Renovation Status, Total Sales by Years Since Renovation

7. Results

7.1 Output Screenshots

Figure 1: Text Table — Count / Average Sale Price / Basement Area

Area of the House from Basement (in Sqft)	38,643,798
Count of Transformed_Housing_Data2.csv	21,609
Avg. Sale Price	511,619

Figure 2: Total Sales by Years Since Renovation (Histogram)

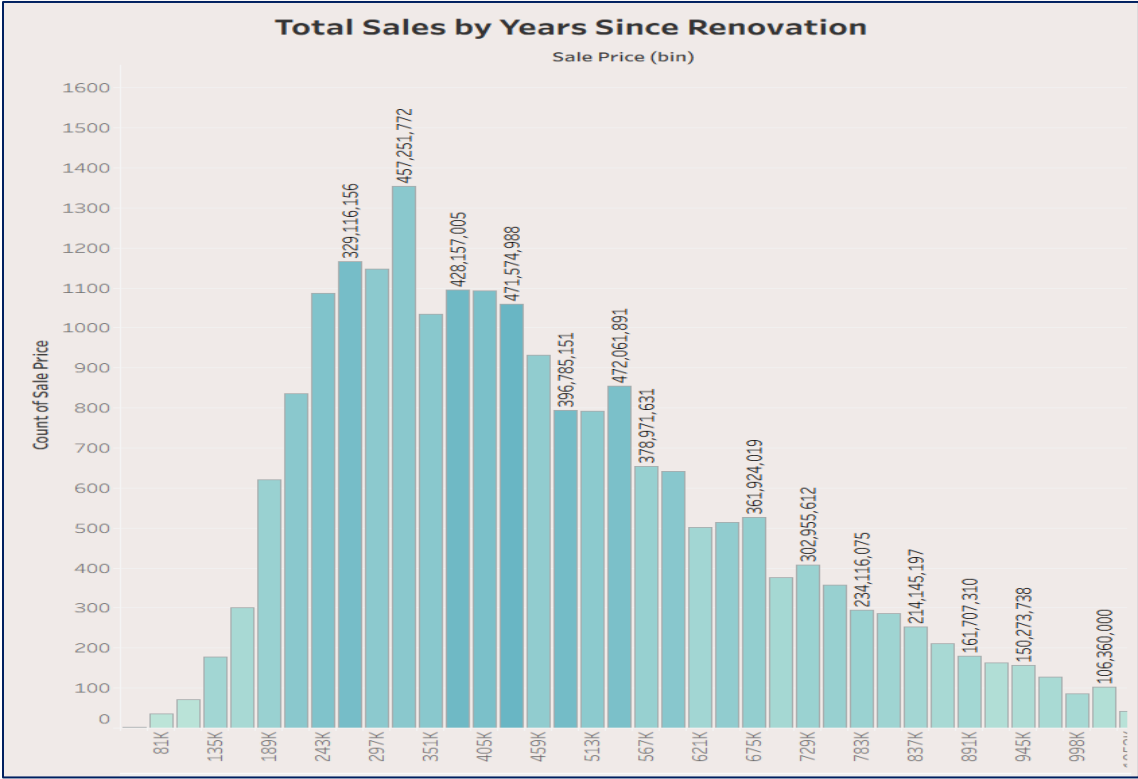
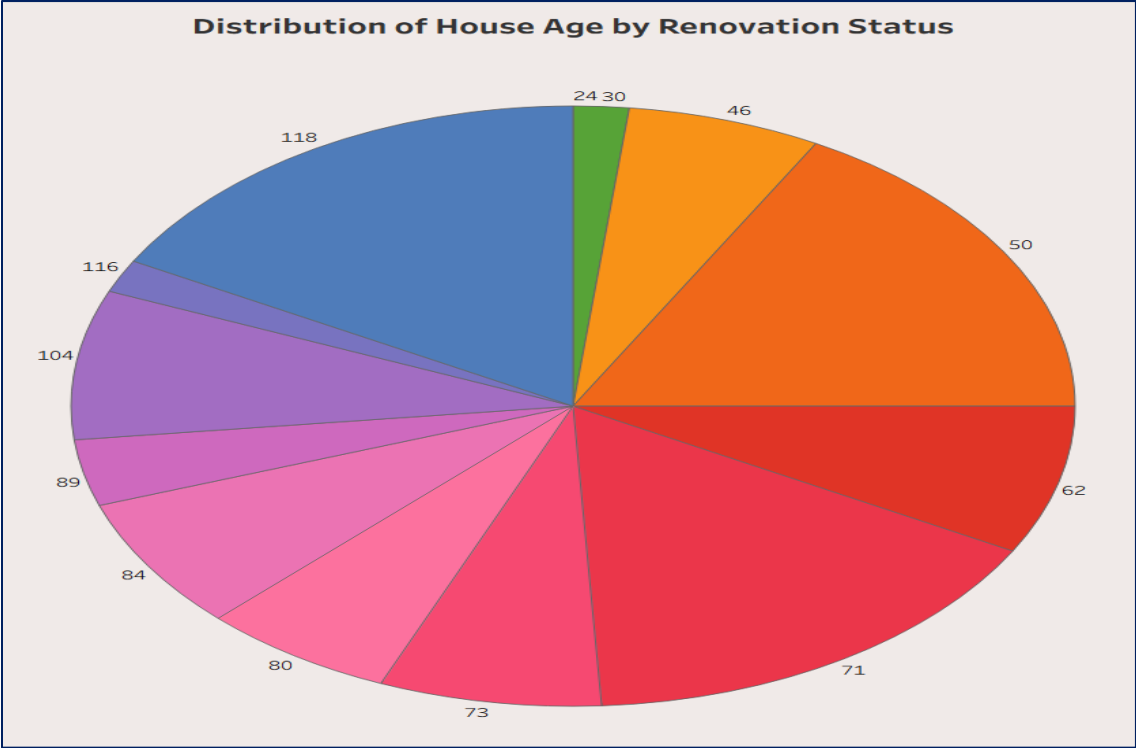


Figure 3: Distribution of House Age by Renovation Status (Pie Chart)



4: House Age Distribution by Bathrooms/Bedrooms/Floors (Bar Chart)

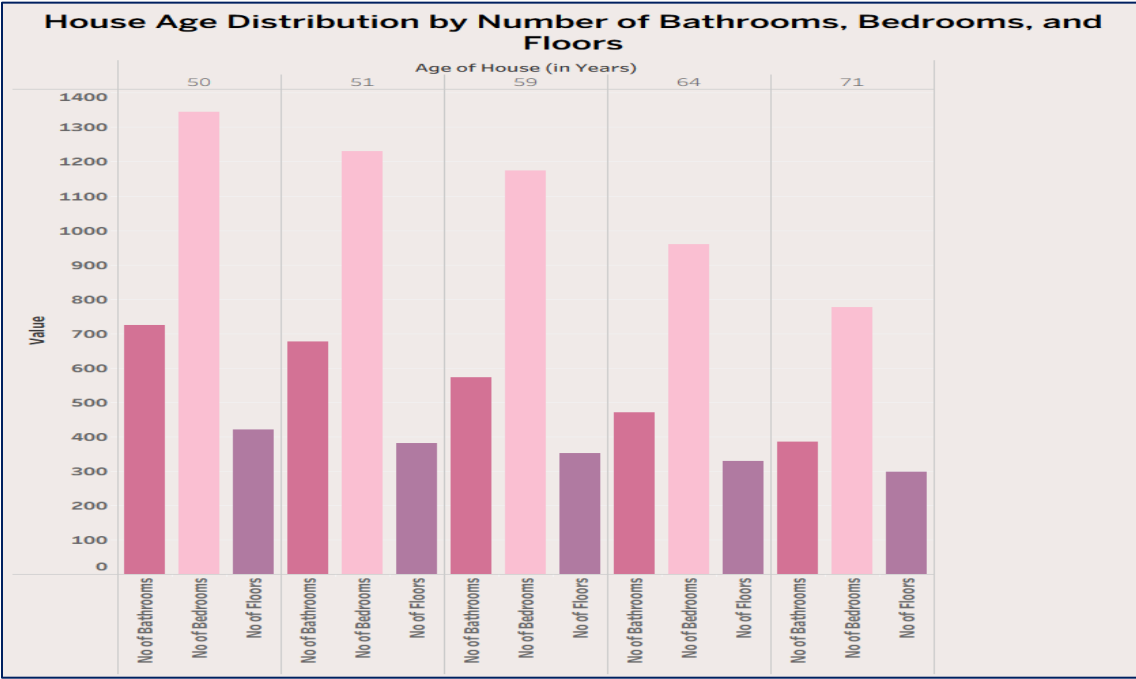


Figure 5: Comprehensive House Data Analysis Dashboard

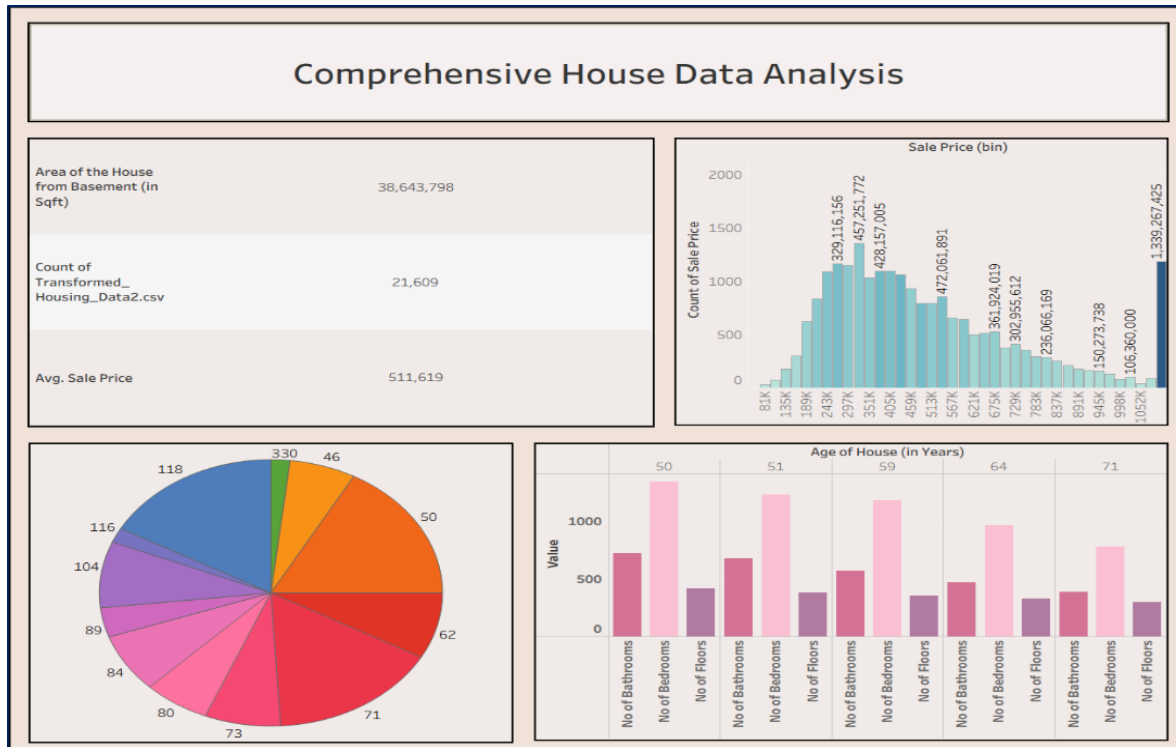
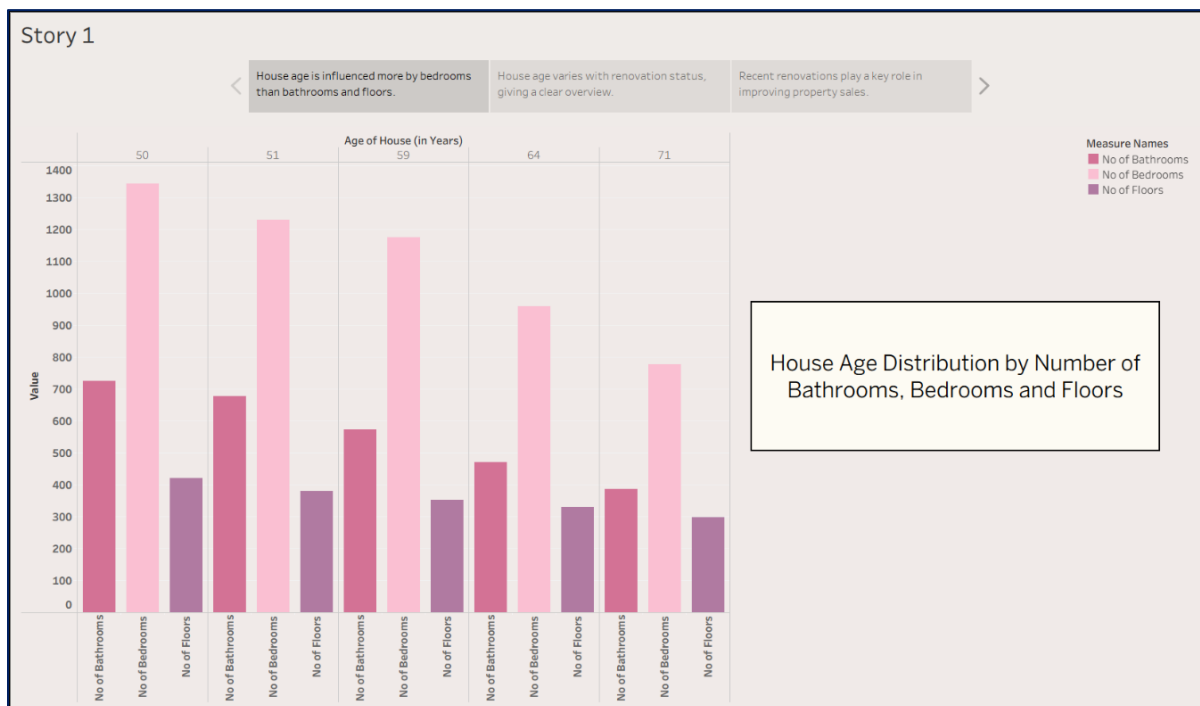


Figure 6: Storyboard — Story Points 1 to 3

Story Point 1 – House Age Distribution by Number of Bathrooms, Bedrooms and Floors

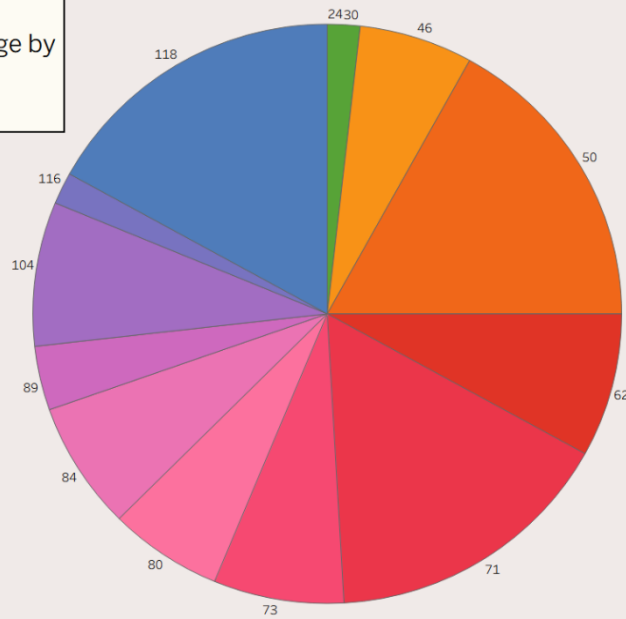


Story Point 2 – Distribution of House Age by Renovation Status

Story 1

< House age is influenced more by bedrooms than bathrooms and floors. House age varies with renovation status, giving a clear overview. Recent renovations play a key role in improving property sales. >

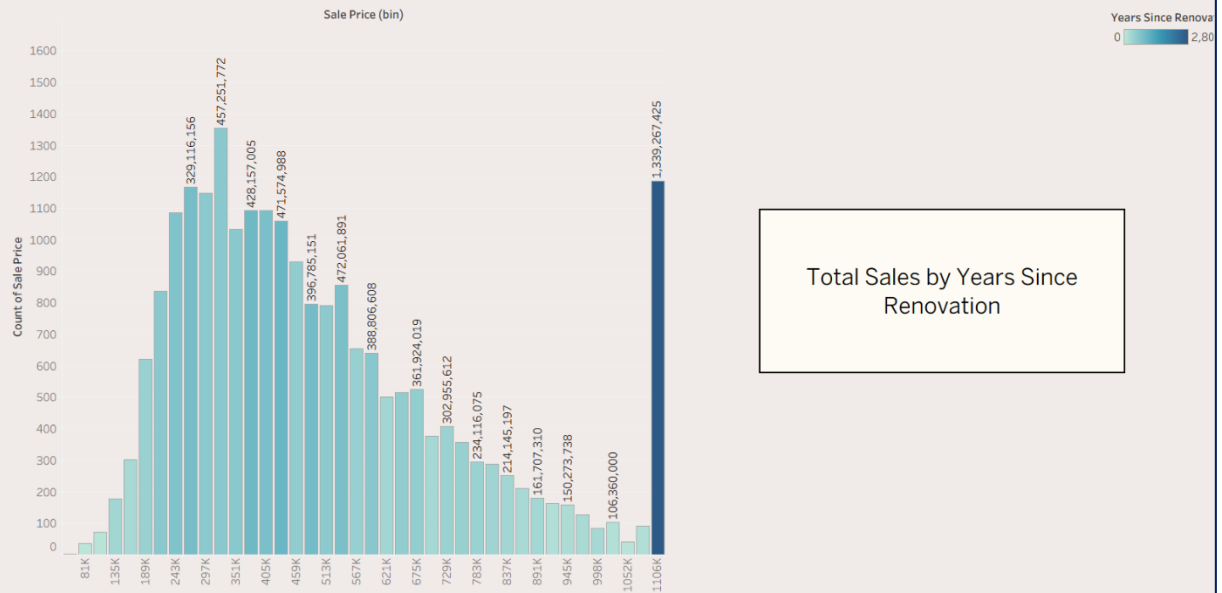
Distribution of House Age by Renovation Status



Story Point 3 – Total Sales by Years Since Renovation

Story 1

< House age is influenced more by bedrooms than bathrooms and floors. House age varies with renovation status, giving a clear overview. Recent renovations play a key role in improving property sales. >



Total Sales by Years Since Renovation

Tableau Public Links:

Dashboard:

https://public.tableau.com/views/Housing_Market_Analysis_17840604208870/Dashboard1?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Storyboard:

https://public.tableau.com/views/Housing_Market_Analysis_17840604208870/Story1?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

8. Advantages & Disadvantages

Advantages

- Consolidates a large, hard-to-interpret dataset (~21,600 records) into a single, easy-to-navigate visual dashboard.
- Combines multiple related dimensions — renovation history, house age, room configuration, and basement area — into one cohesive view instead of isolated reports.
- The guided storyboard makes insights accessible to non-technical viewers without requiring them to interpret each chart independently.
- Published free via Tableau Public, so it is accessible to anyone with a browser and an internet connection, with no dedicated hosting server required.
- The modular workbook structure allows new sheets, filters, or a larger dataset to be added without redesigning the existing dashboard.

Disadvantages

- The dataset is static (a CSV snapshot); it does not reflect real-time market changes unless manually refreshed and re-published.
- The location-based map view (average sale price by zipcode) was deferred due to time constraints, leaving a geographic gap in the current analysis.
- Tableau Public hosts dashboards publicly, so the workbook is not suited to data that must remain private or confidential.
- Users unfamiliar with the dashboard may be unsure where to start among the four visualizations without additional onboarding guidance.

9. Conclusion

This project successfully transformed a raw housing dataset of approximately 21,609 records into an interactive Tableau dashboard and storyboard that helps home buyers and real estate analysts understand what drives housing sale prices. Through structured data cleaning, four purpose-built visualizations, a consolidated 2x2 dashboard, and a three-point guided story, the workbook turns a large, unstructured dataset into clear, actionable insight — reducing the guesswork and manual cross-referencing that both target user groups previously relied on. Both planned sprints were delivered on schedule, and the final workbook was published to Tableau Public and documented end-to-end on GitHub alongside a demonstration video.

10. Future Scope

- Add the deferred map view (average sale price by zipcode) to bring a geographic dimension into the dashboard.
- Extend the workbook to other counties or regions beyond the current King County-style dataset.
- Connect the dashboard to a live data source (e.g., an MLS feed) instead of a static CSV, to support real-time updates.
- Introduce zipcode-level drill-downs and predictive price estimates as part of a potential premium tier for real estate agencies and investors.
- Add default filter presets and a short "how to read this dashboard" note to reduce onboarding friction for first-time viewers.

11. Appendix

Source Code

This project was built entirely within Tableau Desktop/Public and does not involve custom application source code. The only artifacts are the Tableau workbook (.twbx) and the underlying dataset.

Dataset Link

<https://www.kaggle.com/datasets/rituparnaghosh18/transformed-housing-data-2>

GitHub & Project Demo Link

Github Repository Link:

<https://github.com/nityasrialamanda06/housing-market-tableau-project>

Project Demonstration Link:

https://drive.google.com/file/d/1MwJtZoQnZ_mKvKm85ousbra-FF14GzON/view